

Amarm	50219	AgSp1.1	glycine	rm1	1	PGTTPGTITGPDGKP	TKFIVPYGAFTTPGSIPGPDGRPIH	VEPAGPGTTTPGAE	TGPDGKVN		KIYLP	TPK..GP	GGP	75				
Amarm	50219	AgSp1.1	glycine	rm1	2	PGTTPGTVTGP	DKGPK	TKFIVPTGAFTTPGSIPGPDGRPIH	VEPAGPGTTTPGAE	TGPDGKIN		KIYLP	ST..PK	GP	GGP	75		
Amarm	50219	AgSp1.1	glycine	rm1	3								0					
Amarm	50219	AgSp1.1	glycine	rm1	8	PGTTPGTVTGP	DKGPK	IKFIVPTGAFTTPGSIPGPDGRPIH	VEPAGPGTTTPGAE	TGPDGKIN		KIYLP	ST..PK	GP	GGP	75		
Amarm	50219	AgSp1.1	glycine	rm2	1	PGTTPGTETGT	DGKP	NKFVVPKGAFTTPGSIPGPDGKPIH	VEPAGPGTTTPGAE	TGPDGKIN		KIIVV	PTT	TK..APT	GP	75		
Amarm	50219	AgSp1.1	glycine	rm2	78								0					
Amarm	50219	AgSp1.1	glycine	rm2	89								0					
Lcorn	18104	AgSp1.1	glycine	rm1	1	PGTTPGTVTGP	DKGPK	IKFIVPTGAFTTPGSIPGPDGKPIH	VEPAGPGTTTPGAQ	TGPDGKIN		KIIVL	PTT	..PK	AP	QGP	75	
Lcorn	18104	AgSp1.1	glycine	rm1	37	.GTTT	PGTVTGP	DKGPK	TKFIVPKGAFTTPGSIPGPDGKPIH	VEPAGPGTTTPGAE	TGPDGKIN		KLIVV	PTT	PK..GP	QGP	74	
Lcorn	18104	AgSp1.1	glycine	rm1	38	PGTTPGTVTGP	DKGPK	NKFIVPKGAFTTPGSIPGPDGKPIH	VEPAGPGTTTPGAQ	TGPDGKIN		KLIVV	PTT	..PK	GP	QGS	75	
Lcorn	18104	AgSp1.1	glycine	rm1	39	PGTTPGTVTGP	DKGPK	TKFIVPKGAFTTPGSIPGPDGKPIH	VEPAG				45					
Ncruc	58532	and13532	AgSp1	glycine	rm 1	PGTTPGTVTGP	DKGPK	TKFVVPLGAFTTPGSIPGPDGQPIH	VEPAGPGTTTPGAE	TGPDGKVN		KIYLP	PTT	..KG	PGG	74		
Ncruc	58532	and13532	AgSp1	glycine	rm 2	PGTTPGTVTGP	DKGPK	TKFVVPLGAFTTPGSIPGPDGQPIH	VEPAGPGTTTPGAE	TGPDGKVN		KIYLP	PTT	..KG	PGG	74		
Ncruc	58532	and13532	AgSp1	glycine	rm 3	PGTTPGTVTGP	DKGPK	TKFVVPLGAFTTPGSIPGPDGQPIH	VEPAGPGTTTPGAE	TGPDGKVN		KIYLP	PTT	..KG	PGG	74		
Ncruc	58532	and13532	AgSp1	glycine	rm 4	PGTTPGTVTGP	DKGPK	TKFVVPLGAFTTPGSIPGPDGQPIH	VEPAGPGTTTPGAE	TGPDGKVN		KIYLP	PTT	..KG	PGG	74		
Ncruc	58532	and13532	AgSp1	glycine	rm 5	PGTTPGTVTGP	DKGPK	TKFVVPLGAFTTPGSIPGPDGQPIH	VEPAGPGTTTPGAE	TGPDGKVN		KIYLP	PTT	..KG	PGG	74		
Ncruc	58532	and13532	AgSp1	glycine	rm 6	PGTTPGTVTGP	DKGPK	TKFVVPLGAFTTPGSIPGPDGQPIH	VEPAGPGTTTPGVQ	TGPDGK				59				
Mlaby	7048	AgSp1	glycine	rm1b	2	PGTTPGTVTGP	DKGPK	TKFIVPTGAFTTPGSIPGPDGKPIR	VEPAGPGTTTPGAE	TGPDGKIN		KIIVV	PTT	PK	KEP		72	
Mlaby	7048	AgSp1	glycine	rm1a	2	PGTTPGTVTGP	DKGPK	SKYIVPLGAYTTPGSIPGPDGKPIH	VQ	PAGPGTTTPGT	IDSQCNVD		KIILP	TT	PK..G	QQ	73	
Mlaby	7048	AgSp1	glycine	rm1a	4	PGTTPGVVTGP	DGQ	VQFIVPQGAFTTPGSIPGNCKRIH	VK	PAGPGTTTPGAK				53				
Mlaby	7048	AgSp1	glycine	rm1a	13	PGTTPGVVTGP	GGE	LEFVVPKGAFTTPGSIPGNCKPIH	VGP	SGPGTTTPGAK	TDS	DGNID		SIIVL	PTT	PK	DSGPGF	76
Mlaby	7048	AgSp1	glycine	rm2a	21								9					
Mlaby	7048	AgSp1	glycine	rm2a	22	PGST	TPGTVTGP	DGM	P	K				16				
Mlaby	7048	AgSp1	glycine	rm2b	29								23					
Mhut	1271	AgSp1	glycine	rm1	1	PGTTPGTITGPDGKP	IQFIVPYGAFSTPGSIPGPDGPIIL	VEPAESGKT	TPG	IETGPDGKIY		KIYLP	PIIQKGP	DKHRT	77			
Mhut	1271	AgSp1	glycine	rm1	2	PGTTPGTITGPDGRP	TKFIVPYGAFTTPGTIPGPDGKPIQ	VEPAGPGTTTPG	VERGD	DGKVY		KIYLP	PLMHISP	GELI	76			
Mhut	1271	AgSp1	glycine	rm1	3	PGTTPGTITGPDGKP	TKFIVPYGAFTTPGSIPGPDGKPIH	VEPAGPGTTTPG	VKR	GPDGKVY		KIYLP	PTT	PK..GP	GGP	75		
Mhut	1271	AgSp1	glycine	rm1	4	PGTTPGTITGSDGRP	IQFIIPYGAYSTPGSIPGPDGTPIH	VEPAGPGTTTPG	IETGPDGKVY		KIYLP	PTT	PK..GP	GGP	75			
Mhut	1271	AgSp1	glycine	rm1	5	PGTTPGTITGSDGRP	IKFIIPYGAYSTPGSIPGPDGTPIH	VEPAGPGTTTPG	IETGPDGKVY		KIYLP	PT	RP	PK....RP	72			
Mhut	1271	AgSp1	glycine	rm1	6	PGTTPGTITGSDGRP	IKFIIPYGAYSTPGSIPGPDGTPIH	VEPAGPGTTTPG	VKTGPDGKVY		KIYLP	PTT	PK..GP	GGP	75			
Mhut	1271	AgSp1	glycine	rm1	7	PGTTPGTITGSDGRP	IKFIIPYGAFSTPGSIPGPDGTPIH	VEPAGPGTTTPG	VKTGPDGKVY		KIYLP	PTT	PK..GP	GGP	75			
Mhut	1271	AgSp1	glycine	rm1	8	PGST	PGTITGPDGRP	TQFIIPFGAFTTPGSIPGPDGPIH	VEPAG				45					
Mhut	1271	AgSp1	glycine	rm3	17	PGTTPGVYVTD	PDGQA	VKFIVPEDAFITPGYIPGNCKPVVR	VGP	PAG				45				
Aarg	AgSp1	FC511	glycine	rm1	1	PGTTPGTITGPDGKP	TKFIITPLGAFTTPGSIPGPDGRPIH	VQ	PAGPGTTTPGAE	TGADGKIN		KIIV	PTT	PKS		71		
Aarg	AgSp1	FC511	glycine	rm1	2	PGTTPGTVTGP	DKGPK	KKFVVPKGAFTTPGSIPGPDGKPIH	VEPAGPGTTTPGAQ	TGPDGKIN		KIIVV	PTT	TTPKGP	VGP	77		
Aarg	AgSp1	FC511	glycine	rm1	3	PGTTPGTVTGS	SDGKP	KKFIVPK					22					
Aarg	AgSp1	FC511	glycine	rm1	13								55					
Aarg	AgSp1	FC511	glycine	rm1	15	PGTTPGTVTGP	DKGPK	KKFVVPQGAFTTPGSIPGPDGNPIP	VEPAGPGTTTPGTQ	TGPDGKIN		K		62				
Aarg	AgSp1	FC511	glycine	rm2	6	PGTTPGTITGPDGNP	TKFIVPLGAFTTPGSIPGPDGKPIH	VQ	PAGPGTTTPGAE	TGPDGKIN		KIILP	TT	PKRPSYP	..	75		
Aarg	AgSp1	FC511	glycine	rm2	7	PGTTPGTVTGP	DGQ	VKFIVPQGAFTTPGTITLGDGKPIP	VEPQ				45					
Atri	AgSp1	genomic	glycine	rm1	1	PGTTPGTITGPDGKP	TKFIITPLGAFTTPGSIPGPDGRPIH	VQ	PAGPGTTTPGAE	TGPDGKIN		KIIV	PTT		68			
Atri	AgSp1	genomic	glycine	rm1	2	PGTTPGTVTGP	DKGPK	KKFVLPKGAFTTPGSIPGPDGKPIH	VQ	PAGPGTTTPGAQ	TGPDGKIN		KLIVV	PTT	TTPKGP	VGP	77	
Atri	AgSp1	genomic	glycine	rm1	47								0					
Atri	AgSp1	genomic	glycine	rm2	40								55					
Msag	18528	AgSp1.2	glycine	rm 1	1	PGTTPGTVTGP	DKGPK	R					16					
Msag	18528	AgSp1.2	glycine	rm 16	1								0					
Msag	18528	AgSp1.2	glycine	rm 112	1								0					
Msag	18528	AgSp1.2	glycine	rm 113	1	PGTTPGTVTGP	DKGPK	KKFVVPKGAFTTPGSFSGPDGQPIH	VEPAGPGTTTPGVQ	TGPDGKIN		KVVV	PTT	T..KG	P	QGP	75	
Msag	18528	AgSp1.2	glycine	rm 114	1	PGTTPGTVTGP	DKGPK	KKFVVPKGAFTTPGSFSGPDGQPIH	VEPAGPGTTTPGVQ	TGPDGKIN		KVVV	PTT	TK		70		
Msag	18528	AgSp1.2	glycine	rm 115	1								35					
Varen	6869	AgSp1.2	glycine	rm1	1	PGTTPGTITRHDGKP	KEFIIPKGAFTTPGSFSGPDGQPIY	VEPAGPGTTTPGAI	TGPDGKIY		KIIV	PTT	QK		70			
Varen	6869	AgSp1.1	glycine	rm1	1								57					
Varen	6869	AgSp1.2	glycine	rm1	2	PGTTPGTITGS	DGKP	TKFIIPMGAFTTPGSMPGPDGRPIR	VEPAGPGTTTPGVV	TRHDGKIY		KIIVP	AMQMG	SDE	...	74		
Varen	6869	AgSp1.1	glycine	rm1	2	PGTTPGTVTGR	DGKP	RKFIVPNGAFTTPGSFSGPDGEPIS	VEPAGPGTTTPG	IETDS	DGKIK		KVVV	PTT	PK..G	PEGP	75	
Varen	6869	AgSp1.2	glycine	rm1	3	PGTTPGTVTGK							11					
Varen	6869	AgSp1.2	glycine	rm1	10								6					
Varen	6869	AgSp1.1	glycine	rm1	11								0					
Varen	6869	AgSp1.2	glycine	rm1	11	PGTTPGTVTGK	DGKP	TKFIITPLGAFTTPGSIPGPDGRPIR	VEPAGPGTTTPGVV	TRPDGKIY		KIIV	PTT	IHMGS	DE	...	74	
Varen	6869	AgSp1.2	glycine	rm1	12	PGTTPGTVTCNDGKP	TKFIITPLGAFTTPGSIPGPDGRPIR	VEPAGPGTTTPGVV	TRPDGKIY		KIIV	PTT	IHMGS	DE	...	74		
Varen	6869	AgSp1.2	glycine	rm1	13	PGTTPGTVTGK	DGKP	TKFIITPLGAFTTPGSIPGPDGRPIR	VEPAGPGTTTPGVV	TRPDGKIY		KIIV	PTT	IHMGS	DE	...	74	
Varen	6869	AgSp1.2	glycine	rm1	14	PGTTPGTVTGK	DGKP	IKFIITPLGAFTTPGSIPGPDGRPIR	VEPAG				45					

Amarm	50219	AgSp1.1	glycine	rm1	1	GG.....	QTTPTPVP	GPRGKPIQLPAG	98
Amarm	50219	AgSp1.1	glycine	rm1	2	GG.....	QTTPTPVP	GPR.....	88
Amarm	50219	AgSp1.1	glycine	rm1	3			CKPIQLPAG	10
Amarm	50219	AgSp1.1	glycine	rm1	8	GF.....	NFA..TP.....	ATTPTPIPGPGCKPIQVVPAG	103
Amarm	50219	AgSp1.1	glycine	rm2	1	GG.....	FPLGP...AGFPLGPGGQPI...	YPYGPSPYGPQGPGGKT.TTTPIPGPDGEPISIVPAG	131
Amarm	50219	AgSp1.1	glycine	rm2	78		APTGGGGFPLRT.....	TGPGGQPI...YPYGPSPYGPQGPGGK.....	37
Amarm	50219	AgSp1.1	glycine	rm2	89		APTGGGGFPL.....	PGPGGQPI...YPYGPSPYGPQGPGGK.....	35
Lcorn	18104	AgSp1.1	glycine	rm1	1	GI.....	NPYYPMRPGR.....		87
Lcorn	18104	AgSp1.1	glycine	rm1	37	GL.....	NPYYPMGPEGKPM...	YPYGPSPFPGQGPGGQT.TKTPVP	126
Lcorn	18104	AgSp1.1	glycine	rm1	38	GQ.....	PL.....	FPFGPQ...GPV...GQTTTTPIPGPDCKPIQLVPAG	110
Lcorn	18104	AgSp1.1	glycine	rm1	39				45
Ncruc	58532	and13532	AgSp1	glycine	rm 1		PGFNF.....	GTPATTTPTIQGGCKPIQVVPAG	103
Ncruc	58532	and13532	AgSp1	glycine	rm 2		PGFNF.....	GTPATTTPTIIPGGGCKPIQVVPAG	103
Ncruc	58532	and13532	AgSp1	glycine	rm 3		PGFNF.....	GTPATTTPTIIPGGGCKPIQVVPAG	103
Ncruc	58532	and13532	AgSp1	glycine	rm 4		PGFNF.....	GTPATTTPTIIPGGGCKPIQVVPAG	103
Ncruc	58532	and13532	AgSp1	glycine	rm 5		PGFNF.....	GTPATTTPTIIPGGGCKPIQVVPAG	103
Ncruc	58532	and13532	AgSp1	glycine	rm 6				59
Mlaby	7048	AgSp1	glycine	rm1b	2		TGPGGQPLNPQ..GPISPFPGGGQSM...	NPLGPGGS..GG...QTTTPEPIPGPNCKPLQIVPAG	128
Mlaby	7048	AgSp1	glycine	rm1a	2	.F.....	Y.....	FP.....VT...PKPTPTPVTGPGCKPIQLIPAG	101
Mlaby	7048	AgSp1	glycine	rm1a	4				53
Mlaby	7048	AgSp1	glycine	rm1a	13		QFPTGPGGQPLNPT..G....	LGPRGQPL.....	91
Mlaby	7048	AgSp1	glycine	rm2a	21		PTGPGGQPLNP.....	LGPRGQPL.....	60
Mlaby	7048	AgSp1	glycine	rm2a	22				16
Mlaby	7048	AgSp1	glycine	rm2b	29				23
Mhut	1271	AgSp1	glycine	rm1	1	K.....		RPSY..THVKDKPIQVVPAG	96
Mhut	1271	AgSp1	glycine	rm1	2			LGKASHGRPIQLIPAG	92
Mhut	1271	AgSp1	glycine	rm1	3	GFYFG.....		MPQYIPGPGGRPIQIVPAG	99
Mhut	1271	AgSp1	glycine	rm1	4	GFYFG.....		MPQYIPGPGGRPIQIVPAG	99
Mhut	1271	AgSp1	glycine	rm1	5	GFYFG.....		MPQYIPGPGGCKPIQLIPAG	96
Mhut	1271	AgSp1	glycine	rm1	6	GFYFG.....		MPQYIPGPGGCKPIQLIPAG	99
Mhut	1271	AgSp1	glycine	rm1	7	GFYFG.....		MPQNIIPGPDCKPIQLVPAG	99
Mhut	1271	AgSp1	glycine	rm1	8				45
Mhut	1271	AgSp1	glycine	rm3	17				45
Aarg	AgSp1	FC511	glycine	rm1	1		PEEPSGRPM...	YPFPGSPYGP...EQTTTTPIPGPDCKPLQIEPAG	114
Aarg	AgSp1	FC511	glycine	rm1	2	GG.....	MPL..SPYYPQGPGGQPM...	NPFAPGSPYGP...EQTTTTPIPGPDCKPLQIEPAG	129
Aarg	AgSp1	FC511	glycine	rm1	3				22
Aarg	AgSp1	FC511	glycine	rm1	13	DG.....	QPMYPS..GP.....	Q...GPG.GQTTTTPIPGPDCKPIQLIPAG	91
Aarg	AgSp1	FC511	glycine	rm1	15				62
Aarg	AgSp1	FC511	glycine	rm2	6			IPTTTTPIPGP.CQPIQLIPAG	96
Aarg	AgSp1	FC511	glycine	rm2	7				45
Atri	AgSp1	genomic	glycine	rm1	1		PKTPQGSQGPM...	YPFPGSPYGP...EQTTTTPIPGPDCKPLQIEPAG	114
Atri	AgSp1	genomic	glycine	rm1	2	GG.....	MPL..SPYSPQGPGGQPM...	YPFPGSPYGP...EQTTTTPIP.....	116
Atri	AgSp1	genomic	glycine	rm1	47				13
Atri	AgSp1	genomic	glycine	rm2	40		KGPLGPGGQPM...	YPSGPQGTG.....GQTTTTPIPGPDGNPIQLIPAG	97
Msag	18528	AgSp1.2	glycine	rm 1					16
Msag	18528	AgSp1.2	glycine	rm 16			GPQGPGGYPLGPGGQ...T.....		16
Msag	18528	AgSp1.2	glycine	rm 112			RGPQGPGGYPLGPGSPFPGKPGGGQTTTPASVQ	GPDGEPIQVEPAG	46
Msag	18528	AgSp1.2	glycine	rm 113	GG.....		YPLGPGSPFPGKPGGGQTTTPASVQ	GPDGEPIQVEPAG	115
Msag	18528	AgSp1.2	glycine	rm 114					70
Msag	18528	AgSp1.2	glycine	rm 115					35
Varen	6869	AgSp1.2	glycine	rm1	1			PETHPACRPIQLVPAG	86
Varen	6869	AgSp1.1	glycine	rm1	1			GPGKMFGPMRPRRQRTTPPSVEGPNCKPIQLIPAG	92
Varen	6869	AgSp1.2	glycine	rm1	2			QGM...NFATTTPIPGPGGRPIQLIPAG	99
Varen	6869	AgSp1.1	glycine	rm1	2	GGY..PMRPR.....			83
Varen	6869	AgSp1.2	glycine	rm1	3				11
Varen	6869	AgSp1.2	glycine	rm1	10			HMGSDQGMNFVTTTPIPGPGGRPIQLIPAG	37
Varen	6869	AgSp1.1	glycine	rm1	11		GPEG.....	PGGYPMGPGNFRPMGPRQRTTPSSVRGPKCKPIQLIPAG	45
Varen	6869	AgSp1.2	glycine	rm1	11			QGM...NFATTTPIPGPGGRPIQLIPAG	99
Varen	6869	AgSp1.2	glycine	rm1	12			QGM...NFATTTPIPGPGGRPIQLIPAG	99
Varen	6869	AgSp1.2	glycine	rm1	13			QGI...NFATTTPIPGPGGRPIQLIPAG	99
Varen	6869	AgSp1.2	glycine	rm1	14				45

 non-conserved
 similar
 $\geq 50\%$ conserved